1.45 Proof

Chapter 7 · Original slide 24 · 4: Proof

Terminology

A conjecture is a mathematical statement that has yet to be proven.

One famous conjecture is Goldbach's Conjecture.

It states "*Every even integer greater than 2 can be expressed as the sum of two primes.*"

It has been verified up to 4×10^{18} (that's big!); this provides evidence that it is true, but does not prove it is true!

A theorem is a mathematical statement that has been proven.

One famous misnomer was Fermat's Last Theorem, which states "*If n is an integer where $n > 2$, then $a^n + b^n = c^n$ has no non-zero integer solutions for a, b, c.*" It was 300 years until this was proven in 1995. Only then was the 'Theorem' in the name then correct!

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A proof must show all assumptions you are using, have a clear sequential list of steps that logically follow, and must cover all possible cases.

You should usually make a concluding statement, e.g. restating the original conjecture that you have proven.

a. Proof by Deduction

This is the simplest type, where you start from known facts and reach the desired conclusion via deductive steps.

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Proof by Deduction requires you to start from known facts and end up at the conclusion. It is not acceptable to start with to the conclusion, and verify it works, because you are assuming the thing you are trying to prove.

Example: Prove that if three consecutive integers are the sides of a right-angled triangle, they must be 3, 4 and 5.

Incorrect Proof:

Let the lengths be 3,4,5.

Therefore:

$3^2 + 4^2 = 5^2$
$25 = 25$

This satisfies Pythagoras' Theorem, and the numbers are consecutive.

Correct Proof:

We are assuming the thing we are trying to prove!

We are only assuming things in the 'if' bit. This is fine!

The underlying problem is that this technique doesn't prove there can't be other consecutive integers that work – we have only verified 3,4,5 is one such solution.

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a. Proof by Deduction

Exam Tip: This is quite a common last parter.

Prove that $x^2 + 4x + 5$ is positive for all values of x.

Anything squared is at least 0. This is formally known as the '*trivial inequality*'.

Test Your Understanding

Prove that the sum of the squares of two consecutive odd numbers is 2 more than a multiple of 8.

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11 Prove that $x^2 + 8x + 20 \geq 4$ for all values of x.

12 The equation $kx^2 + 5kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < \frac{12}{25}$. (4 marks)

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13 The equation $px^2 - 5x - 6 = 0$, where p is a constant, has two distinct real roots. Prove that p satisfies the inequality $p > -\frac{25}{24}$. (4 marks)

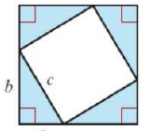
18 A circle has equation $(x - 1)^2 + y^2 = k$, where $k > 0$.
The straight line L with equation $y = ax$ cuts the circle at two distinct points.
Prove that $k > \dfrac{a^2}{1 + a^2}$.

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20 The diagram shows a square and four congruent right-angled triangles.

Use the diagram to prove that $a^2 + b^2 = c^2$.



19 Prove that the line $4y - 3x + 26 = 0$ is a tangent to the circle $(x + 4)^2 + (y - 3)^2 = 100$.

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b. Proof by Exhaustion

This means breaking down the statement into all possible smaller cases, where we prove each individual case.

(This technique is sometimes known as 'case analysis')

Prove that $n^2 + n$ is even for all integers n.

c. Disproof by Counter-Example

While to prove a statement is true, we need to prove every possible case (potentially infinitely many!), we only need one example to disprove a statement.

This is known as a counterexample.

Disprove the statement:

"$n^2 - n + 41$ is prime for all integers n."

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4 Prove that all cube numbers are either a multiple of 9 or 1 more or 1 less than a multiple of 9.

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9 a Prove that for any positive numbers p and q:
$p + q > \sqrt{4pq}$

b Show, by means of a counter-example, that this inequality does not hold when p and q are both negative. (2 marks)